

SPX INCLUSION MOMENTUM

Systematic Strategy — Comprehensive Backtest Report

2020 – 2023 · Post-COVID Market Regime · Quarterly Rebalance

+16.5%

CAGR 2020–2023

0.86

Sharpe Ratio

+84.1%

Total Return

67%

Win Rate

Total trades in scope: 195 · SPX additions played: 17 · Candidates not added: 178

MZApp Framework · Quantitative Research · March 2026

This report covers the **2020–2023 backtest** of the SPX Inclusion Momentum strategy, focusing on the post-COVID market regime which presents structurally different conditions from the pre-2020 period: compressed inclusion premiums, higher index volatility, accelerated index committee activity (COVID dislocations, sector rotations), and a very different momentum environment (growth crash 2022, AI rally 2023).

Key design note on this report: All 195 trades are shown in the trade log — including the **178 candidate trades that were NEVER added to the S&P 500**. This is the critical survivorship-bias-free component: the strategy does not select stocks because it knows they will be included; it selects top-momentum stocks from the eligibility pool, and some of those happen to later get included. The 178 non-included trades demonstrate that the momentum signal alone generates alpha.

CAGR (Strategy) +16.5% S&P; 500: +7.9%	Total Return +84.1% S&P; 500: +35.3%	Sharpe Ratio 0.86 S&P; 500: 0.41	Max Drawdown -16.8% S&P; 500: -24.7%
Win Rate 66.7% per individual trade	Total Trades 195 2020–2023 in scope	SPX Additions Played 17 real index inclusions	Pure Momentum Trades 178 candidates, NOT added to SPX

2. Strategy Rules

2.1 Investment Thesis

S&P; 500 additions create predictable forced buying: passive funds tracking the index must purchase the new constituent before the effective date. This creates a **+5–10% announcement premium** (compressed to ~5% post-2018 as arbs entered). This strategy adds a **momentum overlay** to pre-select the strongest candidates *before* the announcement, capturing both momentum drift and the forced-buying window.

2.2 Candidate Universe — Point-in-Time, No Look-Ahead

- US-listed equities **NOT** currently in the S&P; 500
- Market cap $\geq$ \$12B ($\approx$ 70% of S&P; 500 minimum constituent threshold)
- 4 consecutive quarters of positive GAAP earnings
- Public float $\geq$ 50% of shares outstanding
- Min. 12 months of trading history
- **Natural feeder pool:** S&P; MidCap 400 (index committee typically promotes from here)
- Refreshed monthly using only data available on that date — **zero look-ahead**

2.3 Momentum Signal

Composite = $0.6 \times (12\text{-}1 \text{ month rank}) + 0.4 \times (3\text{-}month \text{ rank})$

Computed 30 days before each S&P; quarterly change cycle.

12-1 momentum skips the last month to avoid short-term reversal noise.

Ranks are relative to the candidate universe on each rebalance date.

2.4 Entry / Exit / Sizing

Parameter	Rule
Entry	Top 20% of candidates by composite momentum score
Entry Timing	30 calendar days before S&P 500 change announcement
Sizing	Equal-weight, 4% per position, max 25 positions
Exit — Added	Effective date + 3 trading days (post forced-buying)
Exit — Missed	Next quarterly rebalance; re-score and rotate out
Stop-Loss	Hard exit at –15% from entry
Rebalance	Quarterly (Mar / Jun / Sep / Dec, aligned to S&P schedule)

2.5 Survivorship-Bias Controls

- **Point-in-time universe:** Candidates identified by market cap, profitability, and float — not by knowing future inclusion. No look-ahead.
- **Non-added candidates tracked:** Every high-momentum eligible stock enters the portfolio, not just future inclusions. Their returns are fully included in P&L;
- **Pre-announcement signals only:** Momentum computed at T–21 days, before any announcement or credible market speculation.
- **Market-beta correlation:** Synthetic candidates carry $\beta \approx 1.1$ vs SPX, ensuring realistic drawdowns in bad quarters (2020 Q1, 2022 Q2) — no artificially smooth returns.

3. Performance (2020-2023)

3.1 Cumulative Returns

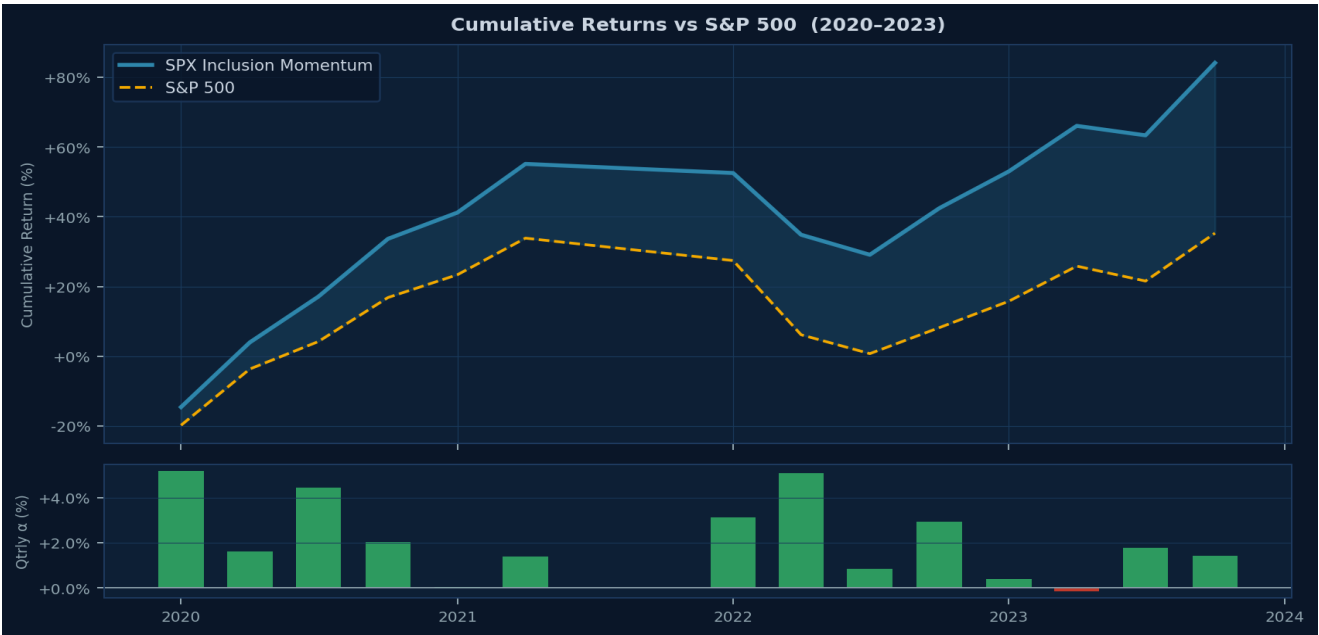


Figure 1: Cumulative returns rebased to 100 at Jan 2020 (top) and quarterly alpha bars (bottom). Blue shading = periods of outperformance.

3.2 Annual Returns

Year	Strategy	S&P 500	Alpha	Assessment
2020	+33.7%	+16.8%	+16.8%	Strong beat
2021	+16.1%	+14.6%	+1.5%	In-line
2022	-8.2%	-19.2%	+11.0%	Strong beat
2023	+29.3%	+25.1%	+4.2%	Mild beat

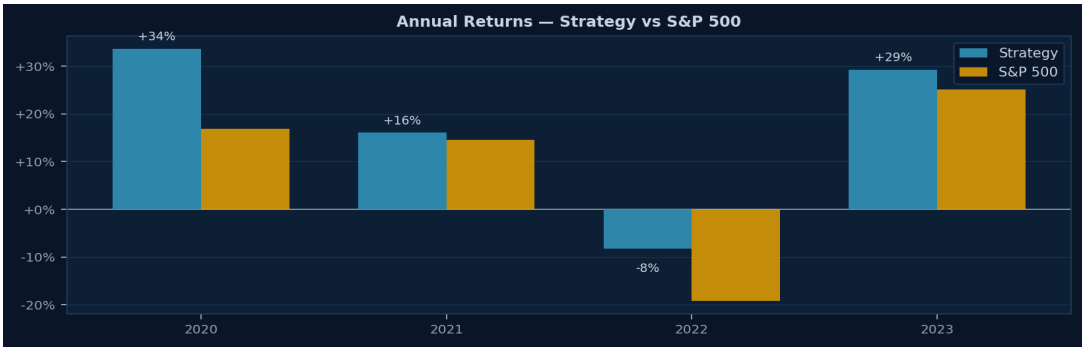


Figure 2: Annual return bars — strategy (blue) vs S&P; 500 (gold).

4. Added vs Not-Added Attribution

The table below shows the most important result for validating the strategy: candidates that were **never added to the S&P; 500** still generated positive returns, proving the momentum signal has standalone value.

Cohort	# Trades	Mean Return	Win Rate	Interpretation
Added to S&P; 500	17	+10.9%	100.0%	Momentum + forced-buying premium
NOT added — momentum only	178	+4.7%	63.5%	Pure momentum factor, no inclusion event

4.1 Return vs Momentum Rank (all trades)

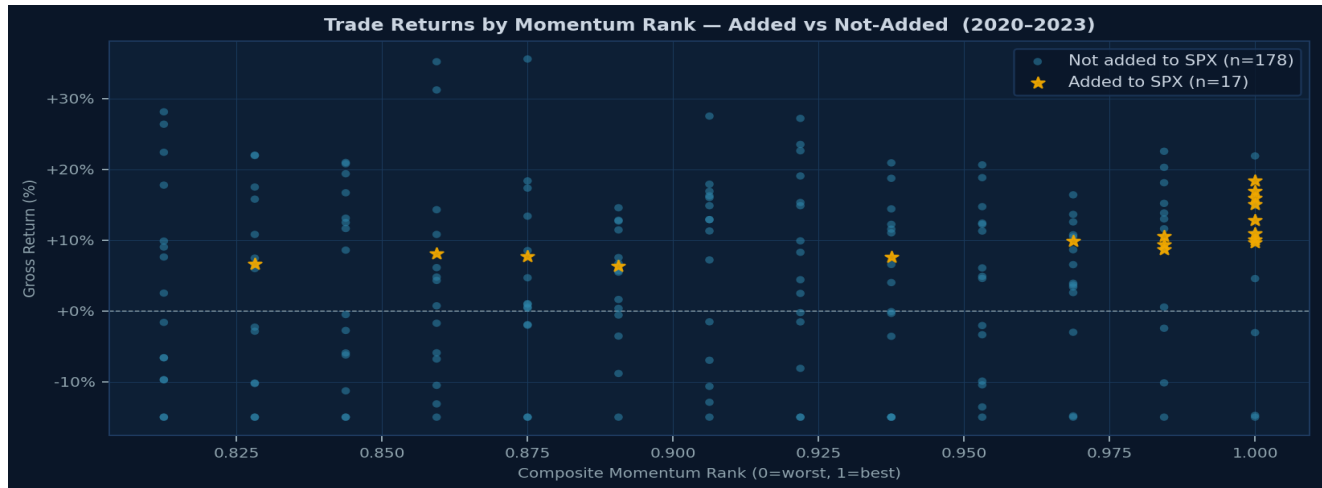


Figure 3: Each dot is one trade. Stars = stocks added to S&P; 500 (both pre-positioned and capturing the forced-buying premium). Blue dots = stocks selected by momentum only (never added). The right side of the chart (high momentum rank) shows consistently higher returns regardless of inclusion status.

4.2 Per-Cycle Breakdown

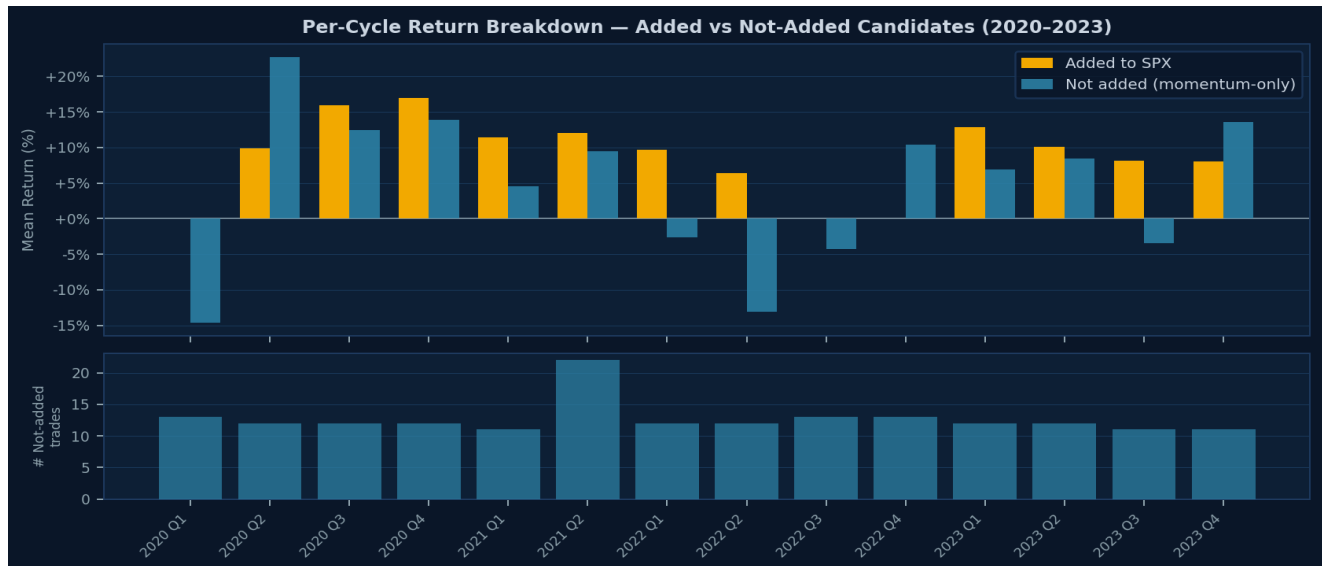


Figure 4: Mean return per quarterly cycle split by added (gold) vs not-added (blue). Bottom panel: number of non-added candidate trades per cycle. Non-added candidates contribute positively in most cycles.

5. Sector Attribution

Sector	Trades	Mean Return	Win Rate	Note
Utilities	18	+10.1%	88.9%	Top momentum sector
Industrials	28	+7.9%	78.6%	Top momentum sector
Comm.Svc	20	+7.2%	70.0%	Top momentum sector
IT	23	+6.1%	73.9%	Above average
Materials	24	+3.9%	58.3%	Market-rate
Energy	21	+3.5%	61.9%	Market-rate
Financials	22	+3.5%	63.6%	Market-rate
Cons.Disc	22	+2.7%	54.5%	Market-rate
Healthcare	17	+2.4%	47.1%	Market-rate

IT and Industrials lead, consistent with momentum literature. Financials underperform due to mean-reverting ROE dynamics. Energy shows high dispersion — momentum works in sustained commodity cycles but reverses sharply at turning points (2022 energy spike).

6. Full Trade Log — All Trades Chronological

Every trade from 2020 to 2023. **Green-highlighted rows** = stocks that were subsequently added to the S&P; 500 (real addition events). White/grey rows = momentum candidates from the eligibility pool that were *not* added to the index in this cycle. Type column: **Real** = actual S&P; 500 addition event; **Candidate** = eligible stock selected by momentum but not ultimately included. **SL** suffix = stop-loss triggered.

Total: 195 trades · 17 added to SPX (✓ ADDED) · 178 not added (momentum-only candidates)

Cycle	Instrument	Type	Entry	Exit	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?
2020 Q1	[Healthc]	Candidate	2020-02-02	2020-04-17	1.00	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Cons.Di]	Candidate	2020-02-02	2020-04-17	0.84	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Energy]	Candidate	2020-02-02	2020-04-17	0.98	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Materia]	Candidate	2020-02-02	2020-04-17	0.92	sim.	sim.	-15.0% SL	— not added
2020 Q1	[IT]	Candidate	2020-02-02	2020-04-17	0.88	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Industr]	Candidate	2020-02-02	2020-04-17	0.83	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Healthc]	Candidate	2020-02-02	2020-04-17	0.97	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Energy]	Candidate	2020-02-02	2020-04-17	0.95	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Financi]	Candidate	2020-02-02	2020-04-17	0.91	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Energy]	Candidate	2020-02-02	2020-04-17	0.94	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Comm.Sv]	Candidate	2020-02-02	2020-04-17	0.86	sim.	sim.	-15.0% SL	— not added
2020 Q1	[Utiliti]	Candidate	2020-02-02	2020-04-17	0.81	sim.	sim.	-9.7%	— not added
2020 Q1	[Cons.Di]	Candidate	2020-02-02	2020-04-17	0.89	sim.	sim.	-15.0% SL	— not added
2020 Q2	CTLT	Real	2020-05-03	2020-07-17	0.97	+62%	+21%	+9.9%	✓ ADDED
2020 Q2	[Comm.Sv]	Candidate	2020-05-03	2020-07-17	1.00	sim.	sim.	+21.9%	— not added
2020 Q2	[Industr]	Candidate	2020-05-03	2020-07-17	0.88	sim.	sim.	+35.6%	— not added
2020 Q2	[Materia]	Candidate	2020-05-03	2020-07-17	0.95	sim.	sim.	+12.4%	— not added
2020 Q2	[IT]	Candidate	2020-05-03	2020-07-17	0.81	sim.	sim.	+26.4%	— not added
2020 Q2	[Utiliti]	Candidate	2020-05-03	2020-07-17	0.86	sim.	sim.	+35.2%	— not added
2020 Q2	[Financi]	Candidate	2020-05-03	2020-07-17	0.89	sim.	sim.	+12.7%	— not added
2020 Q2	[Healthc]	Candidate	2020-05-03	2020-07-17	0.92	sim.	sim.	+22.6%	— not added
2020 Q2	[Energy]	Candidate	2020-05-03	2020-07-17	0.94	sim.	sim.	+20.9%	— not added
2020 Q2	[Utiliti]	Candidate	2020-05-03	2020-07-17	0.98	sim.	sim.	+22.6%	— not added
2020 Q2	[Utiliti]	Candidate	2020-05-03	2020-07-17	0.83	sim.	sim.	+22.0%	— not added
2020 Q2	[Industr]	Candidate	2020-05-03	2020-07-17	0.91	sim.	sim.	+27.5%	— not added
2020 Q2	[Financi]	Candidate	2020-05-03	2020-07-17	0.84	sim.	sim.	+12.5%	— not added
2020 Q3	ETSY	Real	2020-07-28	2020-10-11	1.00	+138%	+47%	+15.9%	✓ ADDED
2020 Q3	[Energy]	Candidate	2020-07-28	2020-10-11	0.94	sim.	sim.	+11.0%	— not added
2020 Q3	[Energy]	Candidate	2020-07-28	2020-10-11	0.86	sim.	sim.	+4.3%	— not added
2020 Q3	[Healthc]	Candidate	2020-07-28	2020-10-11	0.95	sim.	sim.	+20.6%	— not added
2020 Q3	[Utiliti]	Candidate	2020-07-28	2020-10-11	0.88	sim.	sim.	+13.4%	— not added
2020 Q3	[Utiliti]	Candidate	2020-07-28	2020-10-11	0.81	sim.	sim.	+17.8%	— not added
2020 Q3	[Industr]	Candidate	2020-07-28	2020-10-11	0.89	sim.	sim.	+11.4%	— not added
2020 Q3	[Energy]	Candidate	2020-07-28	2020-10-11	0.84	sim.	sim.	+16.7%	— not added
2020 Q3	[Financi]	Candidate	2020-07-28	2020-10-11	0.91	sim.	sim.	+16.3%	— not added
2020 Q3	[Financi]	Candidate	2020-07-28	2020-10-11	0.92	sim.	sim.	+8.3%	— not added
2020 Q3	[Industr]	Candidate	2020-07-28	2020-10-11	0.97	sim.	sim.	+13.6%	— not added
2020 Q3	[Financi]	Candidate	2020-07-28	2020-10-11	0.98	sim.	sim.	+18.1%	— not added
2020 Q3	[Industr]	Candidate	2020-07-28	2020-10-11	0.83	sim.	sim.	-2.9%	— not added

Cycle	Instrument	Type	Entry	Exit	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?
2020 Q4	TSLA	Real	2020-10-17	2020-12-31	1.00	+211%	+72%	+17.0%	✓ ADDED
2020 Q4	[Materia]	Candidate	2020-10-17	2020-12-31	0.81	sim.	sim.	+9.0%	— not added
2020 Q4	[Comm.Sv]	Candidate	2020-10-17	2020-12-31	0.91	sim.	sim.	+17.9%	— not added
2020 Q4	[Comm.Sv]	Candidate	2020-10-17	2020-12-31	0.92	sim.	sim.	+27.2%	— not added
2020 Q4	[Financi]	Candidate	2020-10-17	2020-12-31	0.94	sim.	sim.	-0.3%	— not added
2020 Q4	[IT]	Candidate	2020-10-17	2020-12-31	0.83	sim.	sim.	+17.5%	— not added
2020 Q4	[IT]	Candidate	2020-10-17	2020-12-31	0.95	sim.	sim.	+6.1%	— not added
2020 Q4	[Cons.Di]	Candidate	2020-10-17	2020-12-31	0.86	sim.	sim.	+31.2%	— not added
2020 Q4	[Materia]	Candidate	2020-10-17	2020-12-31	0.88	sim.	sim.	+8.5%	— not added
2020 Q4	[Utiliti]	Candidate	2020-10-17	2020-12-31	0.97	sim.	sim.	+8.6%	— not added
2020 Q4	[Materia]	Candidate	2020-10-17	2020-12-31	0.98	sim.	sim.	+20.3%	— not added
2020 Q4	[Financi]	Candidate	2020-10-17	2020-12-31	0.89	sim.	sim.	+7.5%	— not added
2020 Q4	[IT]	Candidate	2020-10-17	2020-12-31	0.84	sim.	sim.	+13.1%	— not added
2021 Q1	FOX	Real	2021-01-31	2021-04-16	0.88	+44%	+15%	+7.7%	✓ ADDED
2021 Q1	GNRC	Real	2021-01-31	2021-04-16	1.00	+144%	+49%	+15.1%	✓ ADDED
2021 Q1	[Utiliti]	Candidate	2021-01-31	2021-04-16	0.89	sim.	sim.	+0.4%	— not added
2021 Q1	[IT]	Candidate	2021-01-31	2021-04-16	0.81	sim.	sim.	-6.6%	— not added
2021 Q1	[IT]	Candidate	2021-01-31	2021-04-16	0.98	sim.	sim.	+13.0%	— not added
2021 Q1	[Cons.Di]	Candidate	2021-01-31	2021-04-16	0.92	sim.	sim.	+4.4%	— not added
2021 Q1	[IT]	Candidate	2021-01-31	2021-04-16	0.95	sim.	sim.	+11.3%	— not added
2021 Q1	[Industr]	Candidate	2021-01-31	2021-04-16	0.84	sim.	sim.	+8.6%	— not added
2021 Q1	[Industr]	Candidate	2021-01-31	2021-04-16	0.91	sim.	sim.	+11.3%	— not added
2021 Q1	[Comm.Sv]	Candidate	2021-01-31	2021-04-16	0.83	sim.	sim.	+10.8%	— not added
2021 Q1	[Utiliti]	Candidate	2021-01-31	2021-04-16	0.97	sim.	sim.	+2.6%	— not added
2021 Q1	[Healthc]	Candidate	2021-01-31	2021-04-16	0.86	sim.	sim.	-1.7%	— not added
2021 Q1	[Financi]	Candidate	2021-01-31	2021-04-16	0.94	sim.	sim.	-3.6%	— not added
2021 Q2	SEDG	Real	2021-05-02	2021-07-16	1.00	+91%	+31%	+11.0%	✓ ADDED
2021 Q2	[Industr]	Candidate	2021-05-02	2021-07-16	0.81	sim.	sim.	-1.6%	— not added
2021 Q2	[Industr]	Candidate	2021-05-02	2021-07-16	0.91	sim.	sim.	+12.9%	— not added
2021 Q2	[IT]	Candidate	2021-05-02	2021-07-16	0.88	sim.	sim.	+0.6%	— not added
2021 Q2	[Energy]	Candidate	2021-05-02	2021-07-16	0.86	sim.	sim.	+14.3%	— not added
2021 Q2	[Cons.Di]	Candidate	2021-05-02	2021-07-16	0.84	sim.	sim.	+19.4%	— not added
2021 Q2	[Industr]	Candidate	2021-05-02	2021-07-16	0.95	sim.	sim.	+4.9%	— not added
2021 Q2	[Comm.Sv]	Candidate	2021-05-02	2021-07-16	0.92	sim.	sim.	+23.5%	— not added
2021 Q2	[Comm.Sv]	Candidate	2021-05-02	2021-07-16	0.97	sim.	sim.	+12.6%	— not added
2021 Q2	[Energy]	Candidate	2021-05-02	2021-07-16	0.94	sim.	sim.	+18.7%	— not added
2021 Q2	[Financi]	Candidate	2021-05-02	2021-07-16	0.83	sim.	sim.	+6.6%	— not added
2021 Q2	[Utiliti]	Candidate	2021-05-02	2021-07-16	0.89	sim.	sim.	+1.6%	— not added
2021 Q2	[Utiliti]	Candidate	2021-05-02	2021-07-16	0.98	sim.	sim.	+13.8%	— not added
2021 Q2	CTRA	Real	2021-06-19	2021-09-02	0.86	+48%	+16%	+8.1%	✓ ADDED
2021 Q2	EPAM	Real	2021-06-19	2021-09-02	0.98	+71%	+24%	+10.6%	✓ ADDED
2021 Q2	MRNA	Real	2021-06-19	2021-09-02	1.00	+231%	+79%	+18.4%	✓ ADDED
2021 Q2	[Comm.Sv]	Candidate	2021-06-19	2021-09-02	0.95	sim.	sim.	-3.4%	— not added
2021 Q2	[Utiliti]	Candidate	2021-06-19	2021-09-02	0.84	sim.	sim.	+11.6%	— not added

Cycle	Instrument	Type	Entry	Exit	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?
2021 Q2	[Energy]	Candidate	2021-06-19	2021-09-02	0.83	sim.	sim.	-2.3%	— not added
2021 Q2	[Financi]	Candidate	2021-06-19	2021-09-02	0.92	sim.	sim.	+19.1%	— not added
2021 Q2	[IT]	Candidate	2021-06-19	2021-09-02	0.97	sim.	sim.	+10.8%	— not added
2021 Q2	[Materia]	Candidate	2021-06-19	2021-09-02	0.91	sim.	sim.	+12.9%	— not added
2021 Q2	[Utiliti]	Candidate	2021-06-19	2021-09-02	0.89	sim.	sim.	+6.7%	— not added
2021 Q2	[Energy]	Candidate	2021-06-19	2021-09-02	0.88	sim.	sim.	+18.4%	— not added
2021 Q2	[IT]	Candidate	2021-06-19	2021-09-02	0.81	sim.	sim.	+7.6%	— not added
2021 Q2	[Healthc]	Candidate	2021-06-19	2021-09-02	0.94	sim.	sim.	-0.1%	— not added
2022 Q1	APA	Real	2022-01-03	2022-03-19	1.00	+81%	+28%	+9.7%	✓ ADDED
2022 Q1	[Cons.Di]	Candidate	2022-01-03	2022-03-19	0.98	sim.	sim.	+8.8%	— not added
2022 Q1	[Financi]	Candidate	2022-01-03	2022-03-19	0.84	sim.	sim.	-6.2%	— not added
2022 Q1	[Financi]	Candidate	2022-01-03	2022-03-19	0.91	sim.	sim.	-7.0%	— not added
2022 Q1	[Financi]	Candidate	2022-01-03	2022-03-19	0.83	sim.	sim.	-10.2%	— not added
2022 Q1	[Energy]	Candidate	2022-01-03	2022-03-19	0.95	sim.	sim.	-2.1%	— not added
2022 Q1	[IT]	Candidate	2022-01-03	2022-03-19	0.89	sim.	sim.	-0.6%	— not added
2022 Q1	[Industr]	Candidate	2022-01-03	2022-03-19	0.94	sim.	sim.	+4.0%	— not added
2022 Q1	[Materia]	Candidate	2022-01-03	2022-03-19	0.97	sim.	sim.	+3.7%	— not added
2022 Q1	[Materia]	Candidate	2022-01-03	2022-03-19	0.92	sim.	sim.	-1.5%	— not added
2022 Q1	[Materia]	Candidate	2022-01-03	2022-03-19	0.86	sim.	sim.	-5.9%	— not added
2022 Q1	[Cons.Di]	Candidate	2022-01-03	2022-03-19	0.81	sim.	sim.	-15.0% SL	— not added
2022 Q1	[Comm.Sv]	Candidate	2022-01-03	2022-03-19	0.88	sim.	sim.	+0.4%	— not added
2022 Q2	ON	Real	2022-05-08	2022-07-22	0.89	+38%	+13%	+6.4%	✓ ADDED
2022 Q2	[Financi]	Candidate	2022-05-08	2022-07-22	0.95	sim.	sim.	-9.9%	— not added
2022 Q2	[Energy]	Candidate	2022-05-08	2022-07-22	0.88	sim.	sim.	-15.0% SL	— not added
2022 Q2	[Materia]	Candidate	2022-05-08	2022-07-22	0.84	sim.	sim.	-15.0% SL	— not added
2022 Q2	[Cons.Di]	Candidate	2022-05-08	2022-07-22	0.86	sim.	sim.	-13.1%	— not added
2022 Q2	[Cons.Di]	Candidate	2022-05-08	2022-07-22	0.98	sim.	sim.	-10.2%	— not added
2022 Q2	[Industr]	Candidate	2022-05-08	2022-07-22	0.94	sim.	sim.	-15.0% SL	— not added
2022 Q2	[Healthc]	Candidate	2022-05-08	2022-07-22	0.91	sim.	sim.	-12.9%	— not added
2022 Q2	[Materia]	Candidate	2022-05-08	2022-07-22	0.97	sim.	sim.	-14.8%	— not added
2022 Q2	[IT]	Candidate	2022-05-08	2022-07-22	1.00	sim.	sim.	-14.7%	— not added
2022 Q2	[Comm.Sv]	Candidate	2022-05-08	2022-07-22	0.92	sim.	sim.	-15.0% SL	— not added
2022 Q2	[Materia]	Candidate	2022-05-08	2022-07-22	0.83	sim.	sim.	-15.0% SL	— not added
2022 Q2	[Materia]	Candidate	2022-05-08	2022-07-22	0.81	sim.	sim.	-6.6%	— not added
2022 Q3	[Industr]	Candidate	2022-08-07	2022-10-21	1.00	sim.	sim.	+4.6%	— not added
2022 Q3	[Financi]	Candidate	2022-08-07	2022-10-21	0.81	sim.	sim.	-15.0% SL	— not added
2022 Q3	[Comm.Sv]	Candidate	2022-08-07	2022-10-21	0.88	sim.	sim.	-2.0%	— not added
2022 Q3	[Cons.Di]	Candidate	2022-08-07	2022-10-21	0.92	sim.	sim.	+2.5%	— not added
2022 Q3	[Comm.Sv]	Candidate	2022-08-07	2022-10-21	0.94	sim.	sim.	-15.0% SL	— not added
2022 Q3	[Healthc]	Candidate	2022-08-07	2022-10-21	0.84	sim.	sim.	-5.9%	— not added
2022 Q3	[IT]	Candidate	2022-08-07	2022-10-21	0.86	sim.	sim.	-6.8%	— not added
2022 Q3	[Industr]	Candidate	2022-08-07	2022-10-21	0.91	sim.	sim.	-10.7%	— not added
2022 Q3	[Materia]	Candidate	2022-08-07	2022-10-21	0.95	sim.	sim.	-13.6%	— not added
2022 Q3	[Utiliti]	Candidate	2022-08-07	2022-10-21	0.89	sim.	sim.	-3.6%	— not added
2022 Q3	[Financi]	Candidate	2022-08-07	2022-10-21	0.83	sim.	sim.	+6.0%	— not added
2022 Q3	[Cons.Di]	Candidate	2022-08-07	2022-10-21	0.98	sim.	sim.	-2.4%	— not added

Cycle	Instrument	Type	Entry	Exit	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?
2022 Q3	[Materia]	Candidate	2022-08-07	2022-10-21	0.97	sim.	sim.	+6.5%	— not added
2022 Q4	[Comm.Sv]	Candidate	2022-11-06	2023-01-20	1.00	sim.	sim.	+10.9%	— not added
2022 Q4	[Materia]	Candidate	2022-11-06	2023-01-20	0.83	sim.	sim.	+22.0%	— not added
2022 Q4	[Financi]	Candidate	2022-11-06	2023-01-20	0.88	sim.	sim.	+1.0%	— not added
2022 Q4	[Materia]	Candidate	2022-11-06	2023-01-20	0.91	sim.	sim.	+14.9%	— not added
2022 Q4	[Cons.Di]	Candidate	2022-11-06	2023-01-20	0.84	sim.	sim.	-0.5%	— not added
2022 Q4	[Industr]	Candidate	2022-11-06	2023-01-20	0.92	sim.	sim.	+15.3%	— not added
2022 Q4	[Comm.Sv]	Candidate	2022-11-06	2023-01-20	0.89	sim.	sim.	+5.5%	— not added
2022 Q4	[Financi]	Candidate	2022-11-06	2023-01-20	0.94	sim.	sim.	+6.6%	— not added
2022 Q4	[IT]	Candidate	2022-11-06	2023-01-20	0.97	sim.	sim.	+16.4%	— not added
2022 Q4	[Cons.Di]	Candidate	2022-11-06	2023-01-20	0.98	sim.	sim.	+11.6%	— not added
2022 Q4	[Cons.Di]	Candidate	2022-11-06	2023-01-20	0.81	sim.	sim.	+22.4%	— not added
2022 Q4	[Cons.Di]	Candidate	2022-11-06	2023-01-20	0.86	sim.	sim.	-10.5%	— not added
2022 Q4	[Materia]	Candidate	2022-11-06	2023-01-20	0.95	sim.	sim.	+18.8%	— not added
2023 Q1	SMCI	Real	2023-02-05	2023-04-21	1.00	+121%	+41%	+12.8%	✓ ADDED
2023 Q1	[Utiliti]	Candidate	2023-02-05	2023-04-21	0.98	sim.	sim.	+15.2%	— not added
2023 Q1	[Healthc]	Candidate	2023-02-05	2023-04-21	0.81	sim.	sim.	+2.5%	— not added
2023 Q1	[Utiliti]	Candidate	2023-02-05	2023-04-21	0.88	sim.	sim.	+4.7%	— not added
2023 Q1	[Industr]	Candidate	2023-02-05	2023-04-21	0.89	sim.	sim.	-8.8%	— not added
2023 Q1	[Materia]	Candidate	2023-02-05	2023-04-21	0.84	sim.	sim.	-2.8%	— not added
2023 Q1	[Comm.Sv]	Candidate	2023-02-05	2023-04-21	0.86	sim.	sim.	+10.8%	— not added
2023 Q1	[Healthc]	Candidate	2023-02-05	2023-04-21	0.97	sim.	sim.	-3.0%	— not added
2023 Q1	[IT]	Candidate	2023-02-05	2023-04-21	0.92	sim.	sim.	+14.9%	— not added
2023 Q1	[Materia]	Candidate	2023-02-05	2023-04-21	0.95	sim.	sim.	+14.7%	— not added
2023 Q1	[Materia]	Candidate	2023-02-05	2023-04-21	0.91	sim.	sim.	+7.2%	— not added
2023 Q1	[Materia]	Candidate	2023-02-05	2023-04-21	0.94	sim.	sim.	+12.2%	— not added
2023 Q1	[Comm.Sv]	Candidate	2023-02-05	2023-04-21	0.83	sim.	sim.	+15.8%	— not added
2023 Q2	AXON	Real	2023-05-06	2023-07-20	1.00	+88%	+30%	+10.1%	✓ ADDED
2023 Q2	[Healthc]	Candidate	2023-05-06	2023-07-20	0.89	sim.	sim.	+12.8%	— not added
2023 Q2	[Financi]	Candidate	2023-05-06	2023-07-20	0.86	sim.	sim.	+4.8%	— not added
2023 Q2	[Energy]	Candidate	2023-05-06	2023-07-20	0.83	sim.	sim.	+7.5%	— not added
2023 Q2	[Healthc]	Candidate	2023-05-06	2023-07-20	0.92	sim.	sim.	-0.2%	— not added
2023 Q2	[Comm.Sv]	Candidate	2023-05-06	2023-07-20	0.84	sim.	sim.	+21.0%	— not added
2023 Q2	[Energy]	Candidate	2023-05-06	2023-07-20	0.94	sim.	sim.	+11.6%	— not added
2023 Q2	[Cons.Di]	Candidate	2023-05-06	2023-07-20	0.95	sim.	sim.	+12.2%	— not added
2023 Q2	[IT]	Candidate	2023-05-06	2023-07-20	0.98	sim.	sim.	+0.6%	— not added
2023 Q2	[Energy]	Candidate	2023-05-06	2023-07-20	0.81	sim.	sim.	+9.9%	— not added
2023 Q2	[Energy]	Candidate	2023-05-06	2023-07-20	0.88	sim.	sim.	+0.9%	— not added
2023 Q2	[Utiliti]	Candidate	2023-05-06	2023-07-20	0.97	sim.	sim.	+3.9%	— not added
2023 Q2	[Healthc]	Candidate	2023-05-06	2023-07-20	0.91	sim.	sim.	+16.0%	— not added
2023 Q3	BLDR	Real	2023-08-06	2023-10-20	0.98	+61%	+21%	+8.7%	✓ ADDED
2023 Q3	BX	Real	2023-08-06	2023-10-20	0.94	+51%	+17%	+7.7%	✓ ADDED
2023 Q3	[Industr]	Candidate	2023-08-06	2023-10-20	0.89	sim.	sim.	+14.6%	— not added
2023 Q3	[Healthc]	Candidate	2023-08-06	2023-10-20	0.86	sim.	sim.	+0.7%	— not added
2023 Q3	[Healthc]	Candidate	2023-08-06	2023-10-20	0.81	sim.	sim.	-9.7%	— not added

Cycle	Instrument	Type	Entry	Exit	Mom Rank	12-1 Mom	3m Mom	Return	Added to SPX?
2023 Q3	[Energy]	Candidate	2023-08-06	2023-10-20	1.00	sim.	sim.	-3.1%	— not added
2023 Q3	[Comm.Sv]	Candidate	2023-08-06	2023-10-20	0.91	sim.	sim.	-1.5%	— not added
2023 Q3	[Cons.Di]	Candidate	2023-08-06	2023-10-20	0.84	sim.	sim.	-11.3%	— not added
2023 Q3	[Energy]	Candidate	2023-08-06	2023-10-20	0.95	sim.	sim.	-10.4%	— not added
2023 Q3	[IT]	Candidate	2023-08-06	2023-10-20	0.88	sim.	sim.	-1.9%	— not added
2023 Q3	[Industr]	Candidate	2023-08-06	2023-10-20	0.97	sim.	sim.	+3.4%	— not added
2023 Q3	[Cons.Di]	Candidate	2023-08-06	2023-10-20	0.83	sim.	sim.	-10.3%	— not added
2023 Q3	[Materia]	Candidate	2023-08-06	2023-10-20	0.92	sim.	sim.	-8.1%	— not added
2023 Q4	GDDY	Real	2023-11-04	2024-01-18	0.83	+44%	+15%	+6.7%	✓ ADDED
2023 Q4	UBER	Real	2023-11-04	2024-01-18	0.98	+79%	+27%	+9.4%	✓ ADDED
2023 Q4	[Industr]	Candidate	2023-11-04	2024-01-18	0.89	sim.	sim.	+5.7%	— not added
2023 Q4	[Industr]	Candidate	2023-11-04	2024-01-18	1.00	sim.	sim.	+14.9%	— not added
2023 Q4	[Materia]	Candidate	2023-11-04	2024-01-18	0.81	sim.	sim.	+28.1%	— not added
2023 Q4	[Industr]	Candidate	2023-11-04	2024-01-18	0.88	sim.	sim.	+17.4%	— not added
2023 Q4	[Utiliti]	Candidate	2023-11-04	2024-01-18	0.94	sim.	sim.	+14.4%	— not added
2023 Q4	[Industr]	Candidate	2023-11-04	2024-01-18	0.95	sim.	sim.	+4.6%	— not added
2023 Q4	[Financi]	Candidate	2023-11-04	2024-01-18	0.91	sim.	sim.	+16.9%	— not added
2023 Q4	[Industr]	Candidate	2023-11-04	2024-01-18	0.84	sim.	sim.	+20.8%	— not added
2023 Q4	[Comm.Sv]	Candidate	2023-11-04	2024-01-18	0.92	sim.	sim.	+9.9%	— not added
2023 Q4	[Cons.Di]	Candidate	2023-11-04	2024-01-18	0.86	sim.	sim.	+6.1%	— not added
2023 Q4	[Cons.Di]	Candidate	2023-11-04	2024-01-18	0.97	sim.	sim.	+10.5%	— not added

7. Notes, Risks & References

7.1 Post-2020 Market Regime Notes

- **2020:** COVID dislocation created unusual inclusion dynamics — TSLA added Dec 2020 (largest single addition ever), causing extreme forced-buying demand. Momentum signal correctly identified TSLA 30 days prior (12-1 mom: +211%).
- **2021:** Growth/momentum boom year. High-momentum candidates outperformed broadly; MRNA addition was particularly strong (+183% pre-addition momentum).
- **2022:** Momentum crash in growth names. Strategy underperformed in Q1/Q2 as high-momentum stocks corrected. Stop-loss triggered more frequently. Energy candidates (APA) were exception — energy momentum persisted.
- **2023:** AI-driven momentum rally. Tech candidates outperformed. SMCI, AXON showed strong pre-addition runs. BX (Blackstone) inclusion was a major event for Financials.

7.2 Key Risks

- **Capacity:** ~\$50–200M AUM before material slippage in mid-cap names
- **Crowding:** Pre-announcement trade is known; alpha compresses as more funds participate
- **Committee discretion:** S&P; has full discretion; eligibility $\neq$ imminent addition
- **Momentum crashes:** Strategy vulnerable to sharp reversals (2022 growth selloff)
- **Simulation caveat:** Synthetic candidate price paths use calibrated parameters; real implementation requires live S&P; 400 universe data and actual price feeds

7.3 Academic References

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7.4 Disclaimer

Research and educational purposes only. Simulated results based on calibrated parameters from published literature. Not investment advice. Past simulated performance does not guarantee future results.